

Software Engineering

Lab - 4

Vibe Coding Assignment

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Game: Rock Paper Scissors

Tasks to implement:

Task 1: Fix the inverted outcome evaluation bug

Whenever the player plays a winning hand against the CPU, the game registers it as a loss. In `game_engine.determine_winner()`, the outcome lookup dictionary has inverted results for winning combinations: ("ROCK", "SCISSORS"), ("SCISSORS", "PAPER"), and ("PAPER", "ROCK") are assigned to "CPU" while losing combinations are assigned to "PLAYER". Correct the lookup table mappings so Rock beats Scissors, Scissors beats Paper, and Paper beats Rock.

Task 2: Implement "First to X Wins" match victory state

Currently, the game plays endlessly and keeps accumulating scores without a decisive match winner. Implement a target score ceiling . Once either the player or CPU reaches the target score, transition to a `GAME_OVER` match victory screen declaring the overall champion and prompt the player to press R to reset both scores.

Task 3: Implement dynamic / adaptive AI counter-strategy

The CPU currently picks its move completely at random using `random.choice(self.choices)`. Replace this with an adaptive AI strategy that tracks the player's choice history. If the player frequently favors one move (e.g., throwing Rock 60% of the time), bias the CPU's selection towards the counter-move to create an evolving challenge.

Task 4: Implement hand gesture animated icons or icons reveal

The game displays picks only as simple plain text strings. Enhance `game_engine.render()` by drawing procedural geometric representations or icons for each move (a stone shape for Rock, a sheet outline for Paper, crossing blades for Scissors) with a brief shake animation before revealing both selections